Brazilian E-Commerce Analytics

From Raw Transaction Data to Business Insights

Include:

- PostgreSQL | SQL | Power BI
- Independent Data Analytics Portfolio Project

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Understanding Marketplace Performance

Commercial

Which categories and sellers contribute most to merchandise value?

Customer

How much value comes from repeat customers?

Geography

Where are customers and sellers concentrated?

Delivery

How reliable is the delivery operation?

Customer Experience

How are delivery outcomes associated with review scores?

Objective: Transform historical e-commerce transaction data into reliable metrics and actionable business insights.

Understanding the Data

Brazilian E-Commerce Public Dataset by Olist

99,441

Orders

96,096

Unique Customers

32,951

Products

3,095

Sellers

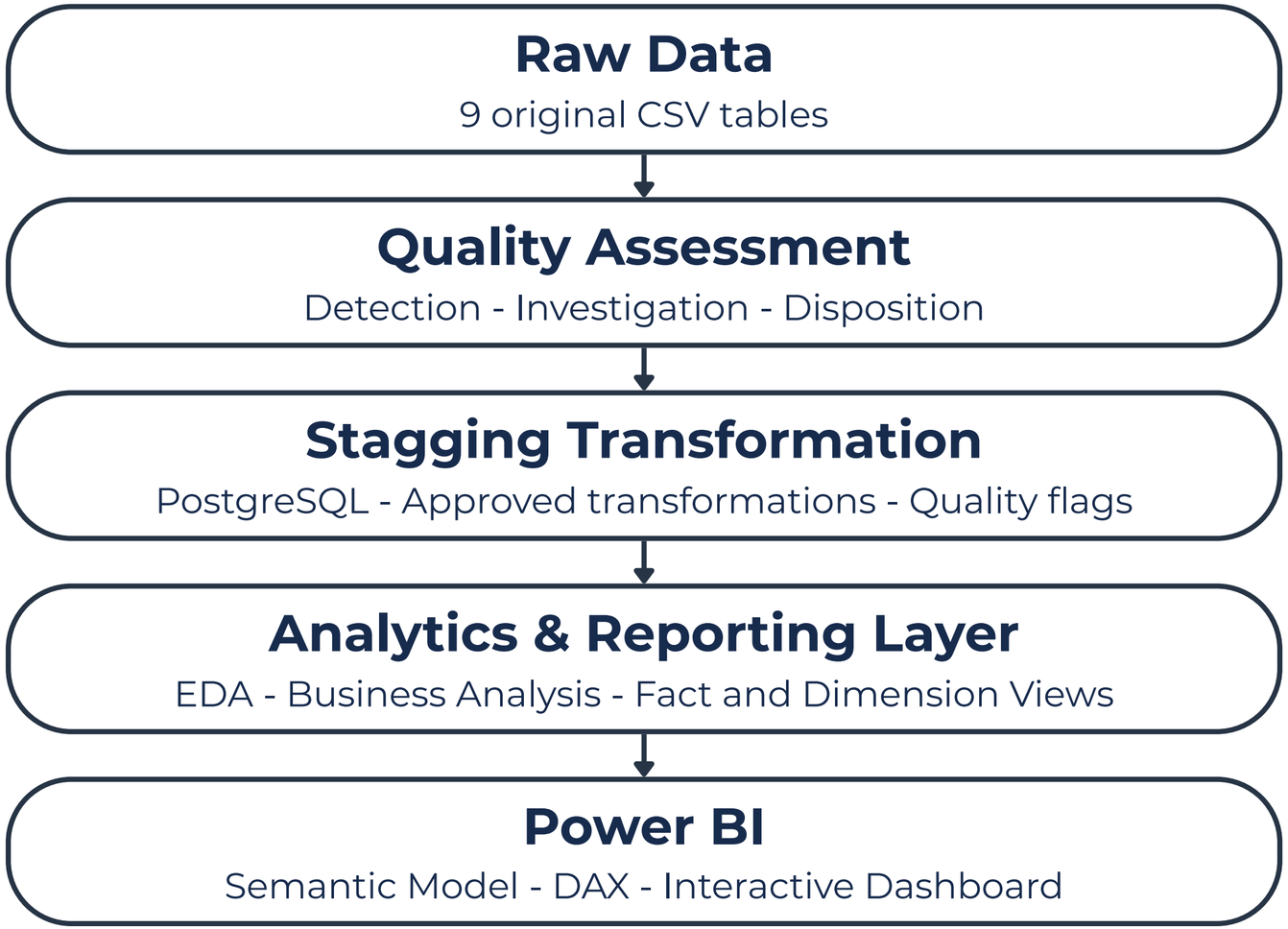
Data Coverage:

- Customers and Sellers
- Orders and Products
- Payments and Reviews
- Geolocation

Historical coverage: September 2016 – October 2018

Source: Olist Brazilian E-Commerce Public Dataset, Kaggle

From Raw Data to Business Intelligence



The analysis separates raw source data, quality-controlled staging objects, and reporting views to maintain traceability and consistent analytical definitions.

PostgreSQL | SQL | Power BI | DAX

Establishing Data Reliability Before Analysis

Finding	Investigation	Treatment
Incomplete product attributes	One incomplete product was referenced by 17 order-item records	Retained and flagged
Duplicate review identifiers	Review identifiers were not unique; repeated reviews had to be interpreted at the correct grain	Defined order-level review aggregation
Missing delivery dates	Eight delivered orders lacked customer delivery dates	Excluded from delivery-performance calculations
Inconsistent order timestamps	Some delivery lifecycle timestamps were out of sequence	Flagged and documented

Raw records were preserved. Records were not automatically deleted simply because they contained missing or inconsistent values.

Preparing Reliable Data for Analysis

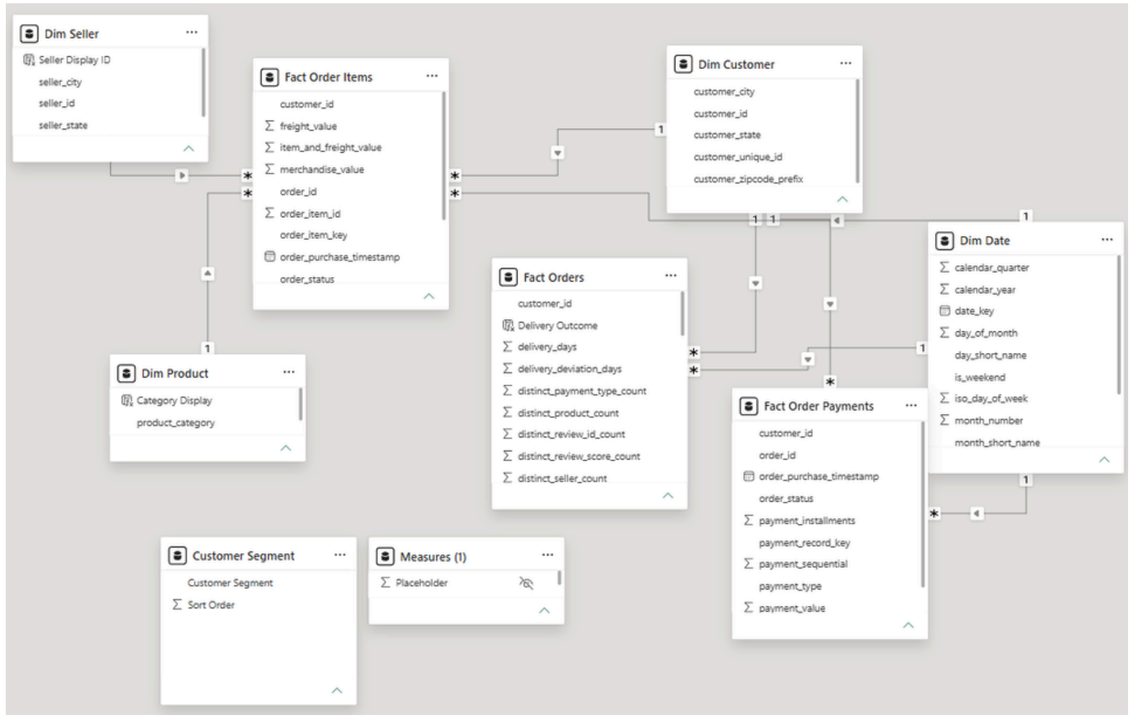
SQL Transformation

Raw Schema

Stagging Schema

Analytics Schema

Power BI Model

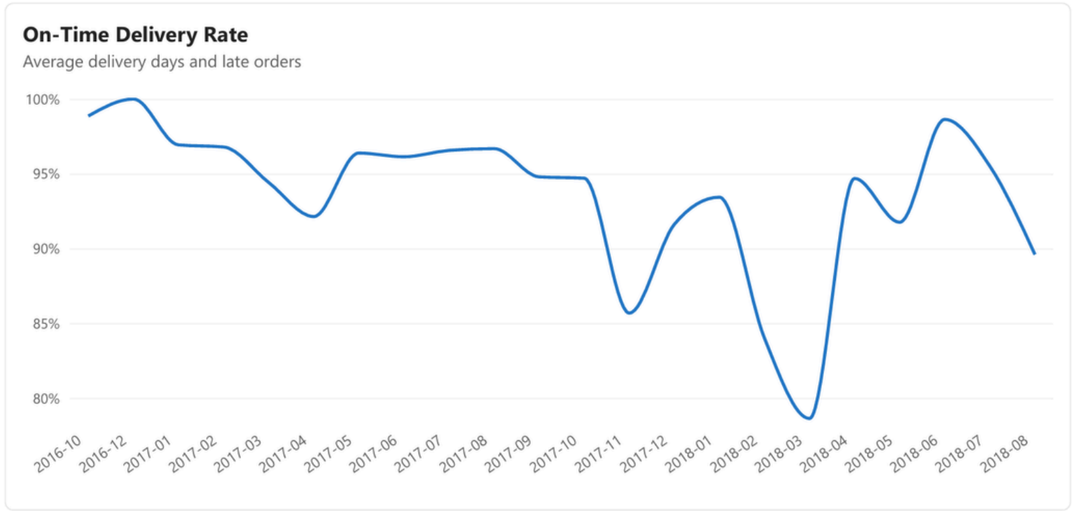


Fact tables remain at their respective transaction grains, with one-to-many, single-direction relationships from dimensions to facts.

Marketplace Performance at a Glance

Total Orders	Delivered Merchandise Value	On-Time Delivery %	Average Delivery Days
99K	$13.22M	91.89%	12.56

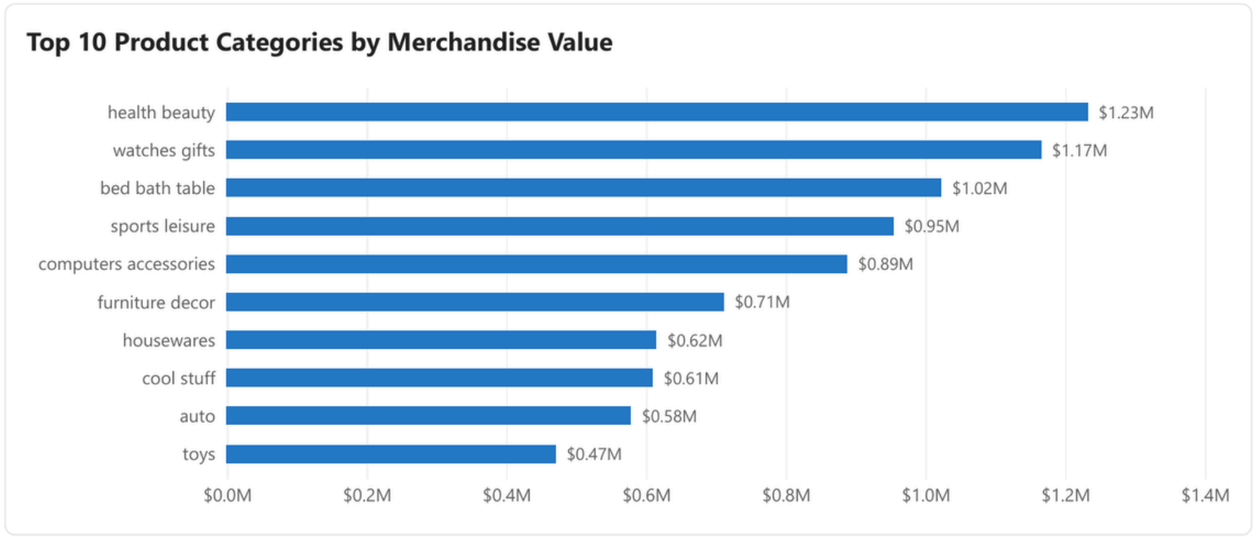


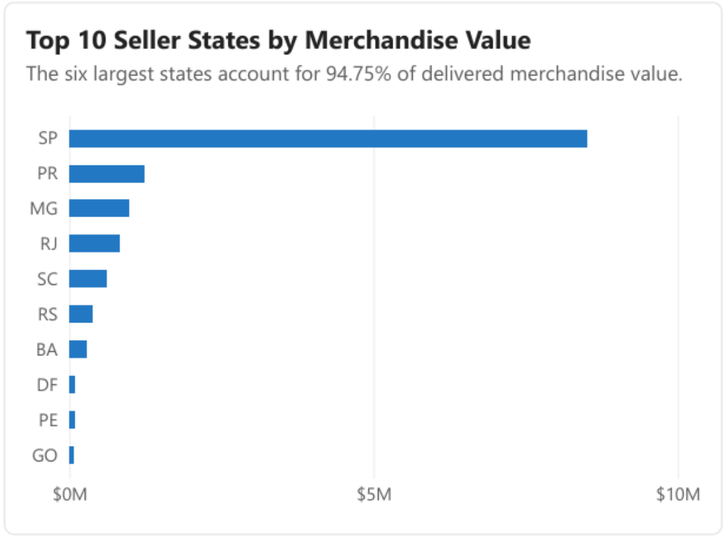


The marketplace recorded substantial transaction activity, while delivery performance provides an important operational dimension for further investigation.

Commercial Contribution Across Categories and Sellers

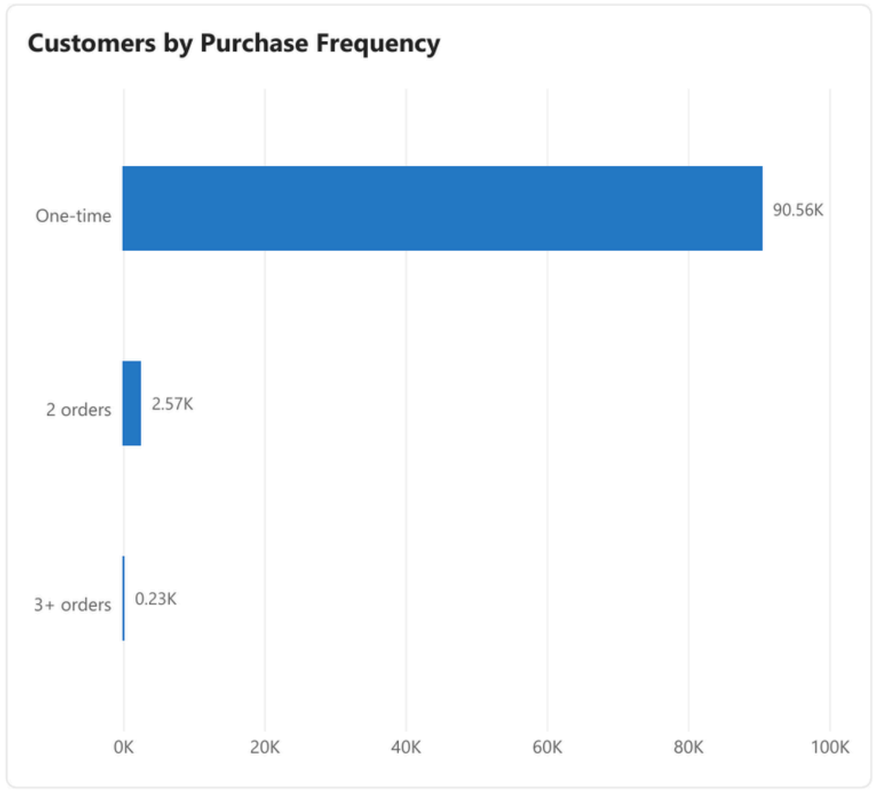
Top 10 Seller Contribution %	Top 50 Seller Contribution %	Top 100 Seller Contribution %
13.27%	33.20%	45.52%





Freight charges vary considerably across product categories, suggesting that category-level shipping characteristics warrant further investigation.

Customer Purchase Frequency and Value



Customers with Delivered Orders
93.36K

Repeat Customers
2.80K

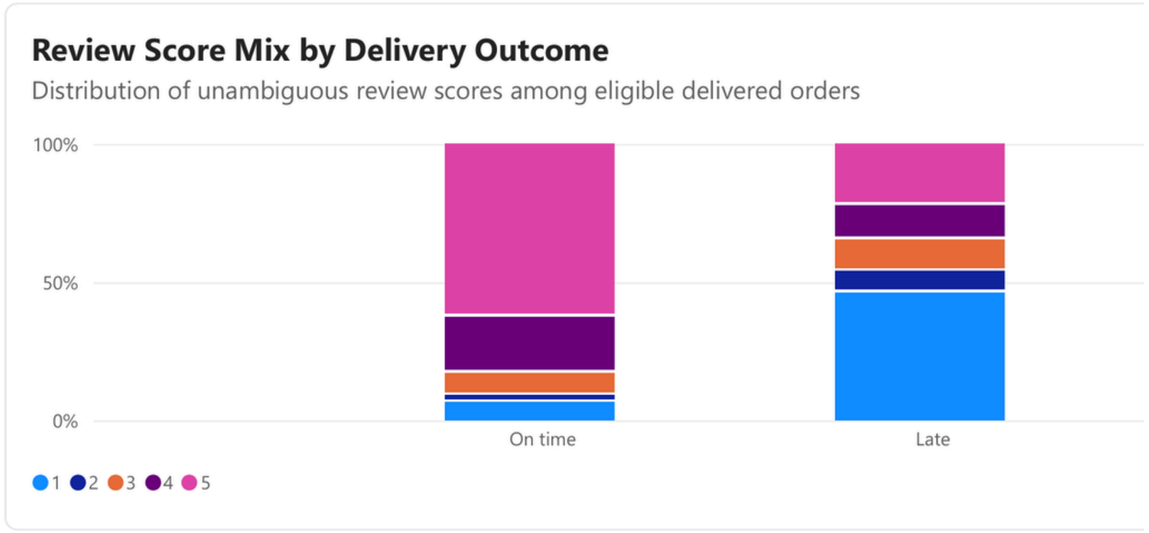
Repeat Customer %
3.00%

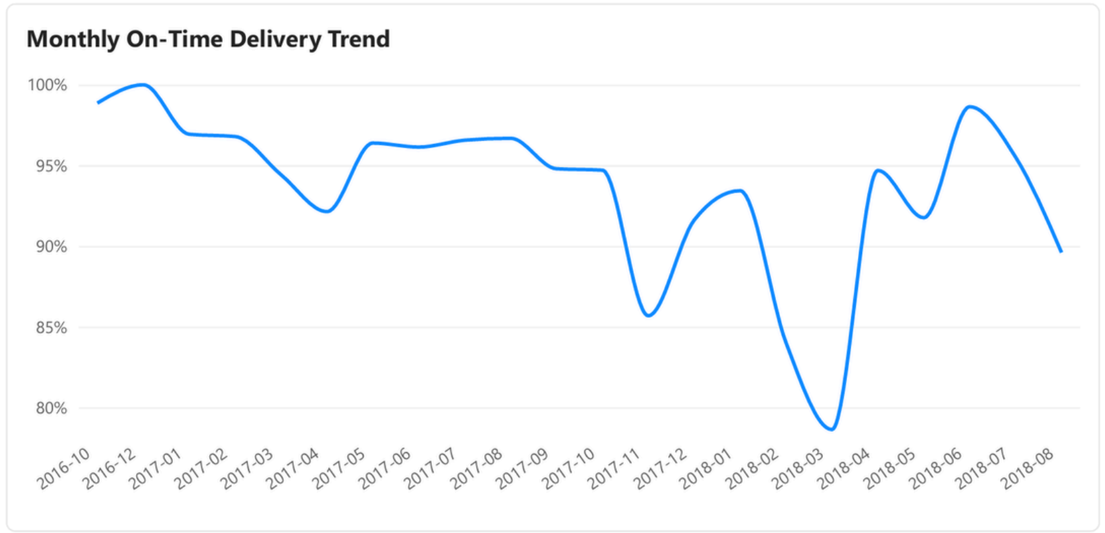
Repeat Customer Merchandise Value Share %
5.51%

Customer segments are based on delivered purchases within the observed dataset period, not lifetime retention or customer lifetime value.

Delivery Reliability and Customer Satisfaction

Eligible Delivered Orders	On-Time Delivery %	Late Orders	Average Delivery Days
96K	91.89%	8K	12.56





Late deliveries are associated with a substantially higher proportion of low customer ratings. That is an observed association, not evidence that delivery delays alone caused lower ratings.

Translating Findings into Business Priorities

Observation	Suggested Investigation
Only 3% of customers made repeat delivered purchases	Investigate repeat-purchase barriers and customer re-engagement opportunities
Top 100 sellers contribute 45.52% of merchandise value	Monitor seller concentration and assess dependence on major contributors
RJ and BA show relatively weak delivery performance at meaningful volumes	Investigate carrier performance, delivery routes, and fulfillment processes
Late deliveries have a much higher low-rating share	Prioritize delivery reliability as an area for customer-experience improvement

From Data Preparation to Decision Support

Technical Impelementation

- PostgreSQL database management
- Data quality assessment and staging transformations
- SQL exploratory and business analysis
- Dimensional reporting layer
- Power BI semantic modeling and DAX

Analytical Output

- Commercial performance insights
- Customer purchasing behavior
- Seller contribution analysis
- Delivery performance and customer satisfaction

Portfolio Deliverables

Github Repository, includes:

- SQL scripts
- Data quality documentation
- Analytical reports
- Power BI report and screenshots



This project demonstrates an end-to-end analytical workflow: transforming raw transactional data into validated reporting datasets and business-oriented insights.